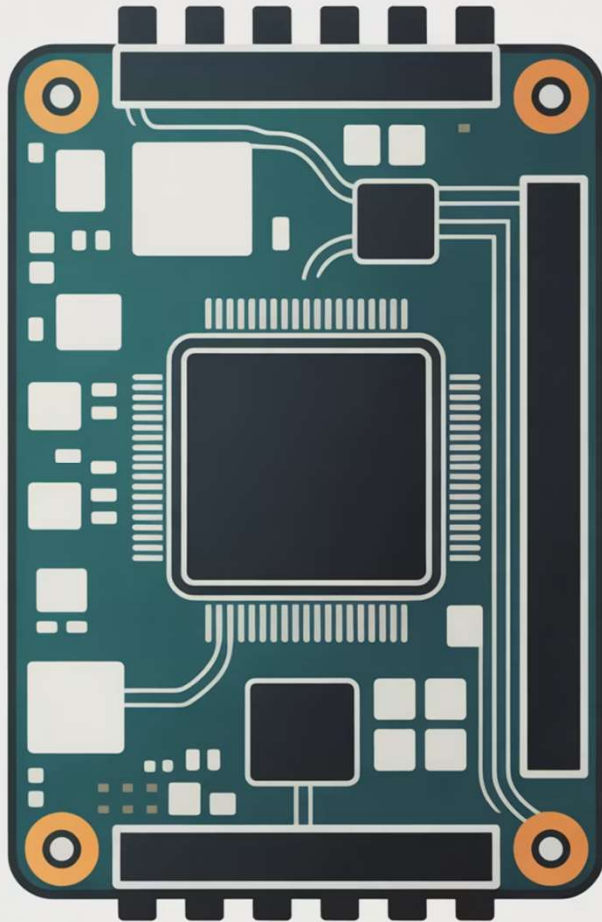




IoT Based SmartPillbox & Vital Health Monitoring

Your Smart Pal for Meds & Monitoring!

This cool system helps you keep track of your meds automatically and lets your folks or doctors check in remotely. Super handy!

From the awesome minds at : E&TC - A

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The Big Problem: Forgetting Your Meds

Forgetting to take medicines is a really big problem in healthcare, especially for older people and those with long-term illnesses like Alzheimer's or diabetes. The World Health Organization says about half of all patients don't take their medicines the way they're supposed to, which can cause serious health problems and make healthcare super expensive.



Brain Blips

People who have trouble remembering things often forget to take their pills.



Oops, Too Many!

Not knowing if you took a pill can make you take too many by mistake, which is really bad!



No Peek-a-boo Monitoring

It's hard for family or nurses to know if someone took their medicine if they're not right there.

Introducing Our Awesome Meds Dispenser!

So, what did we make? For our big final project, we built this super cool automated pill dispenser! It's got some fancy tech inside like super accurate timers, connects to the internet (that's the IoT part!), and sends all the info to a cloud dashboard. Our goal was to help people remember their meds better by giving them pills exactly when they need them and keeping track of everything online.

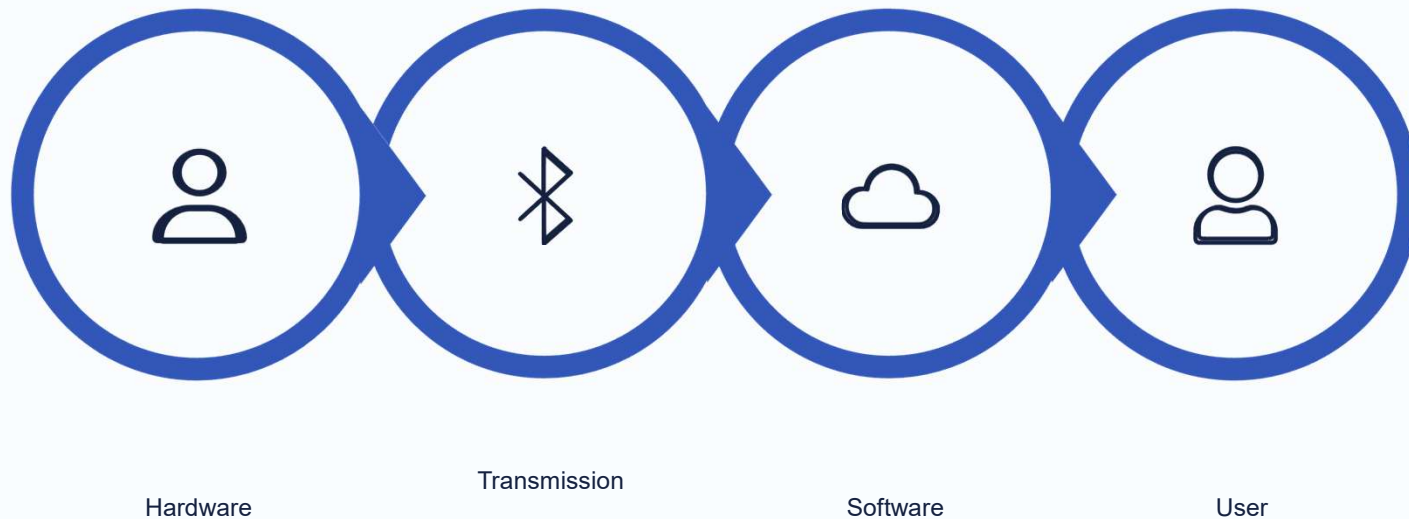
Here's what we focused on when we built it:

- **Totally Automatic:** No more fiddling with pills or trying to remember if you took them. We wanted it to do everything itself!
- **Super Accurate Timing:** We used a special clock (an RTC!) so it dispenses pills at the exact right second, every time.
- **Internet Connected (IoT!):** This means we can set it up, check on it, and get alerts from anywhere using the internet.
- **Easy to Use:** We made sure it has a simple screen right on the dispenser and a handy app for your phone.

We even made a web dashboard called "MedLink IoT Companion" where you can see all your medication info and health stats live!

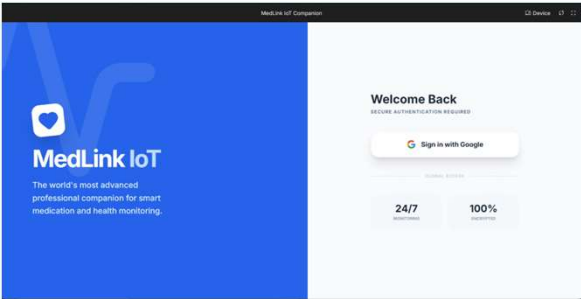


Let's Break Down Our Meds Dispenser!

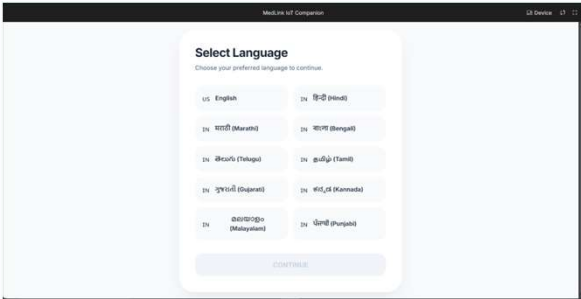


So, how does our awesome meds dispenser actually work behind the scenes? First, our hardware (the actual dispenser with all its cool sensors and motors) gets to work. Then, it sends all the important data wirelessly to our super smart cloud software. Finally, you – whether you're the patient or a caregiver – can easily interact with everything, checking health stats, getting alerts, and managing medication schedules through our dashboard or app!

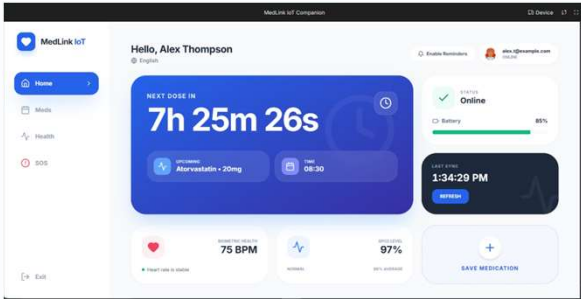
User Interface: Secure Onboarding



Secure Google Sign-in

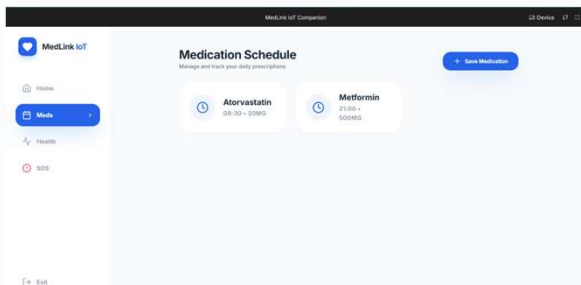


Multi-language Support (including Indian languages)

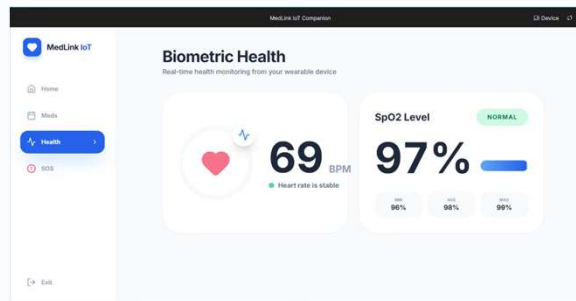


Centralized Home Dashboard with next dose countdown and device status

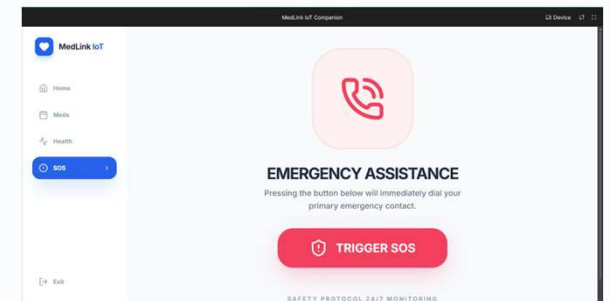
User Interface: Key Features



Medication Schedule: Easy-to-manage list of daily prescriptions



Biometric Health: Real-time visualization of Heart Rate and SpO2 data

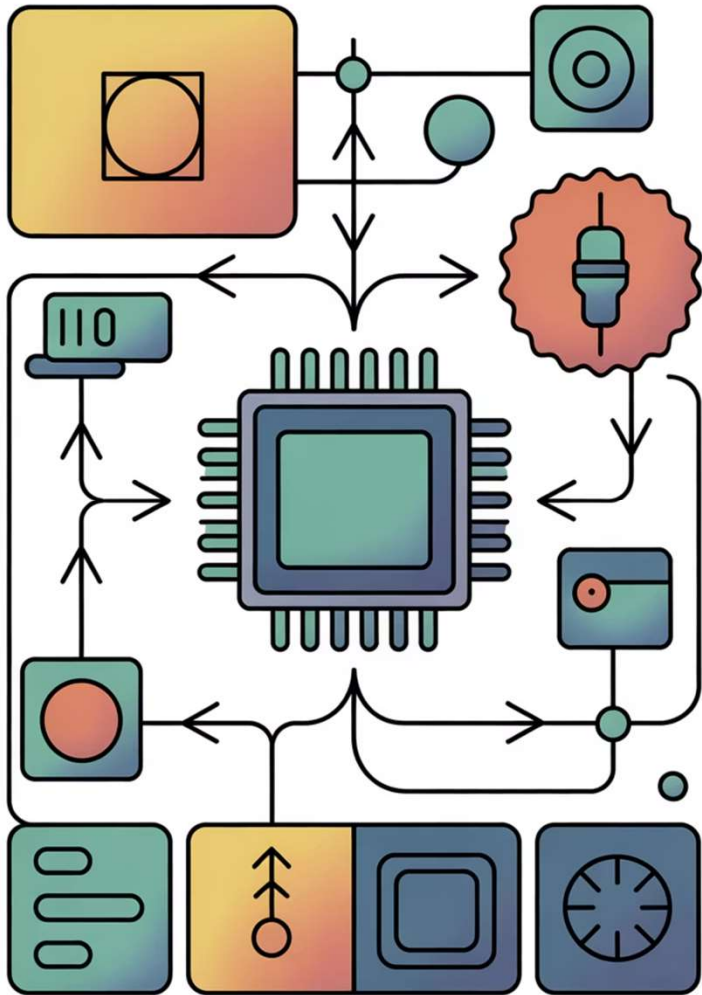


Emergency SOS: One-touch trigger for immediate assistance

Hardware Implementation

Alright, let's dive into the cool gadgets and parts that make up our smart meds dispenser! These are the essential bits and pieces we chose to build our system.

Component	What it Does
Arduino Uno	This is like the mini-computer or "brain" of our project, helping everything run smoothly.
MAX30102 Pulse Sensor	It checks your heartbeat (BPM) right when it happens, so we can keep an eye on your health.
IR Obstacle Sensor	This sensor makes sure that when a pill is supposed to come out, it actually does!
Servo Motor (SG90)	This motor is super important – it's what actually drops the pill out at the right time.
ESP32 Module	This lets our dispenser talk wirelessly to our online dashboard, sending all the important info.
OLED 0.96 inch + I2C	This screen right on the dispenser shows us messages and alerts, so you know what's happening.



Breaking Down Our System's Design

So, how does our project actually work? We designed it to be super clear and efficient, making sure all the parts play nicely together.

Getting Info In (Input Layer)

This is where our system takes in all its commands and data:



- The **MedLink IoT App** on your phone lets you control things remotely.
- Good old **Push Buttons** for direct commands.
- **Wi-Fi** to connect everything!

The Brains (Processing Core)

This is where all the magic happens! Our system's brain:



- The **ESP32 Microcontroller** is the main computer chip doing all the thinking.
- It figures out all the timing and tasks (**Logic Processing**).
- Handles **Wi-Fi Communication** to talk to your phone and the internet.
- Controls the little motor (**Servo Control**) to dispense things.

Showing What Happened (Output Layer)

After all the thinking, our system gives you feedback and takes action:



- A **Servo Motor** moves to dispense items.
- An **LCD Display** shows important messages.
- A colorful **RGB LED** lights up to show status.
- A **Buzzer Alert** makes sounds for notifications!

Basically, this three-part setup lets you tell it what to do, then the ESP32 brain processes your schedule, and finally, it dispenses things with cool lights and sounds!

Our Hardware Choices & Why We Picked Them

We picked all our hardware parts super carefully to make sure they were affordable, dependable, and did exactly what we needed. We also made sure they worked great with our ESP32 brain.

1

ESP32 Microcontroller

What it does: This is like the brain of our project! It handles all the thinking, timing everything, connecting to Wi-Fi, and telling the little motor what to do.

Why we chose this one: We picked the ESP32 because it has two brains (a dual-core processor!), built-in Wi-Fi and Bluetooth, and is super easy to program using the Arduino software. It's perfect for connecting things to the internet!

2

MAX30102 Pulse Sensor Module

What it does: This little guy keeps track of the Heart Beat, and stores the Database

Why we chose this one: We went with the MAX30102 because it's super accurate, and it connects easily to our ESP32. This means our scheduling will always be spot on!

3

SG90 Servo Motor

What it does: This motor is what actually turns the pill drum so the right compartment lines up with the opening to drop the pill.

Why we chose this one: We picked the SG90 because it can turn just enough (180 degrees), has enough power to move the drum, and we can control it precisely. It was just right for making the pill drum spin!

4

0.96 inch OLED Display

What it does: This screen shows us important stuff like the current time, when the next pill is due, if Wi-Fi is connected, and any alerts.

Why we chose this one: We picked this LCD because it's simple to connect, it's big enough to show all our messages, and it doesn't use much battery power. Perfect for keeping you informed!

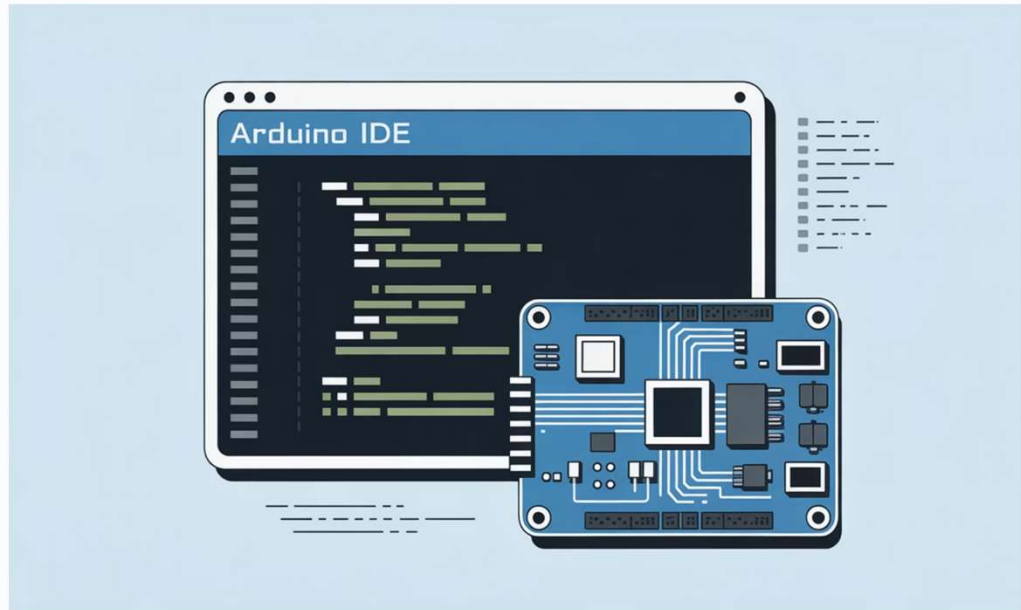
Software Architecture & IoT Integration

MedLink IoT Companion : Our Smart App Brain

MedLink IoT is super cool! It's like a special online toolkit that let us build our own custom app for the pill dispenser without needing to be pro coders. It takes care of all the tricky internet stuff like sending messages to our device (MQTT), keeping things secure (encryption), and even sending notifications to your phone. This means our ESP32 could just focus on dropping pills!

What We Made MedLink IoT Companion Do:

- **Remote Scheduling:** You (or a caregiver) can easily set or change pill times right from a smartphone.
- **Real-Time Notifications:** Get a ping on your phone when it's pill time, or if you miss a dose!
- **Pill History:** It keeps a diary in the cloud of when pills were taken.
- **Talks Both Ways:** Your app can tell the dispenser what to do, and the dispenser can tell your app how it's doing.



How Our Dispenser's Brain Works (Firmware)

What We Coded It In: We wrote the code in C++ using the Arduino IDE, which is a popular platform for hobbyists and students.

Our Main Building Blocks (Libraries):

- `BlynkSimpleEsp32.h` - This helps our ESP32 chat with the Blynk app and the internet.
- `RTCLib.h` - This library lets our ESP32 talk to the time-keeping chip (DS3231) to get the exact time.
- `Servo.h` - This is how we tell the little servo motor to turn and drop the pills.

Basically, our code acts like a tiny manager inside the ESP32. It constantly checks the schedule, makes sure the pills drop on time, and tells the Blynk app what's happening!

Mechanical Design: The Revolver Mechanism

Our pill dispenser has a cool spinning part inside, a bit like a revolver gun, which helps pills fall out smoothly using just gravity. We tried to keep it simple so it works well!

01

The Outside Casing

We 3D-printed a steady, cylinder-shaped outside part. It has just one hole at the bottom where the pills fall out.

02

The Spinning Part Inside

Inside, there's a spinning drum with eight sections, like slices of a pie. Each section holds one pill. It spins around a central pole, moved by a small motor called a servo.

03

Lining Up the Pills

The servo motor turns the drum exactly 45 degrees at a time. This makes sure the right pill section lines up perfectly with the drop hole.

04

Pills Fall Out!

Once a section is lined up, gravity just pulls the pill right through the hole and into a little tray for you.

05

Checking It's Right

Our code keeps track of how much the drum has turned. This way, we always know the right pill section is exactly where it needs to be before a pill drops.



Cool Part About Our Design: Because it's so simple, it doesn't use much battery power, and there are fewer things that can break. This means it works really well every time!

Our Pill Dispenser's Modes & What the Lights Mean

So, our pill dispenser has three main ways it behaves, and it'll always let you know what's up with lights, the screen, and even messages to your phone!



Mode 1: Just Chillin' (Waiting for Pill Time)

LED: A calm blue light

Our system is just patiently watching the clock, waiting for your next scheduled pill.



Mode 2: Time to Take Your Pill!

LED: A super-fast flashing red light

It drops your pill! And if you don't grab it, it'll start buzzing and send alerts to remind you and others.

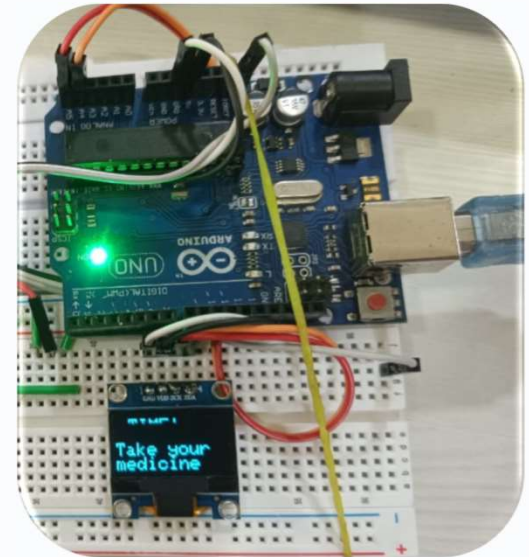
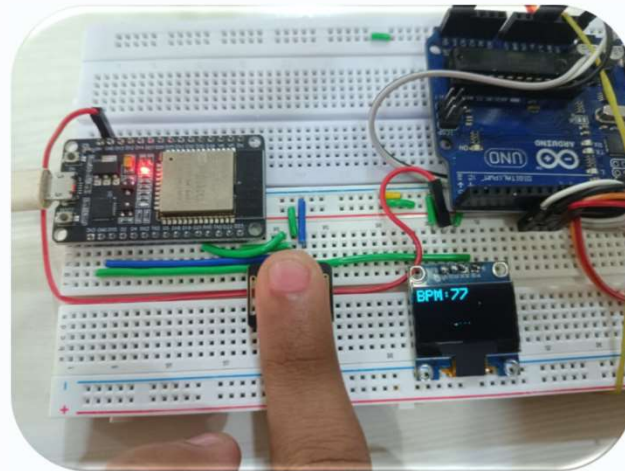
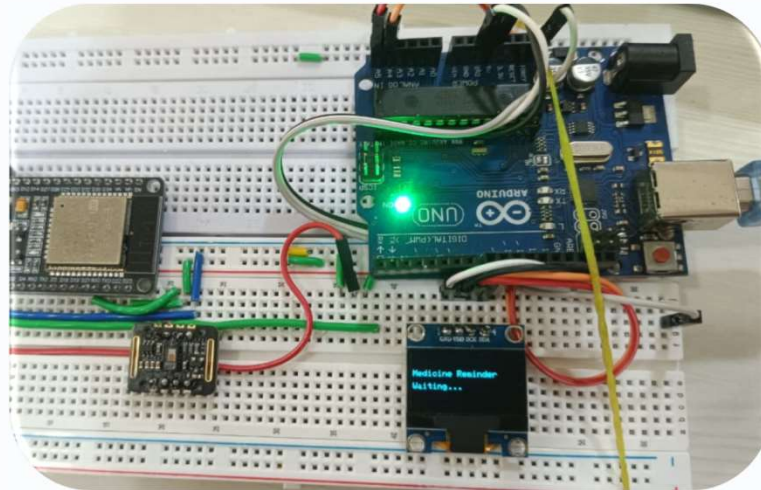
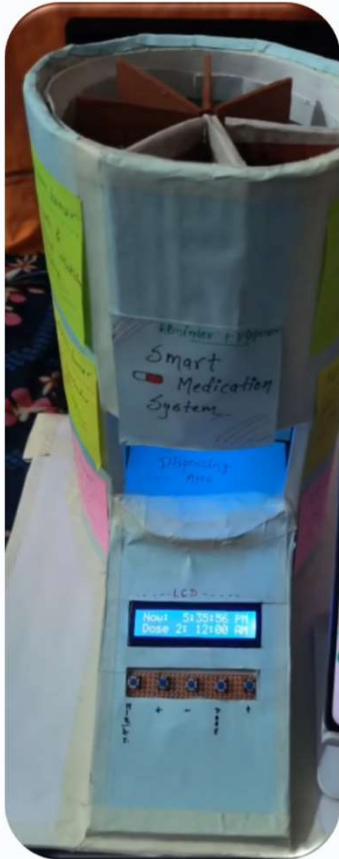


Mode 3: All Good (or Oops!)

LED: A nice steady green light

This means either you took your pill (yay!), or the system noticed you missed it and recorded that for later.

Prototype



What's Awesome About Our Dispenser & What's Next!

Why Our Dispenser Rocks!

Super Affordable

which is way cheaper than other fancy dispensers out there.

Helps People Stay Independent

Grandma and Grandpa can manage their meds easily without someone always watching over them.

Caregivers Can Keep an Eye

Family members get alerts if a dose is missed, so they can quickly check in and help out.

Always on Time

Thanks to our special clock chip, it's accurate to within 2 minutes – super important for meds!

Our Big Ideas for the Future



See If You Took It (with a Camera!)

We want to add a tiny camera to actually *see* if the pill was removed. No more guessing!

Fingerprint Lock

A fingerprint sensor would make sure only the right person can get their pills – super secure!

More Than One Person Can Use It

Imagine if our dispenser could handle different schedules for multiple people, like in a care home!

Talks to Doctors' Records

We could link it to Electronic Health Records so doctors know if you're taking your meds, automatically!

That's a Wrap! (And Thanks!)

So, What Did We Actually Do?

Our project shows how we put together smart tech and internet stuff (like embedded systems and IoT) to help people take their meds on time. It's an affordable solution that helps users be more independent, and gives their families peace of mind because they can check in remotely.

Here's What We're Proud Of:

- We built a fully working prototype with all the main features!
- The software is super tough and handles all sorts of situations without glitches.
- It's really easy to use, with clear sounds and lights to guide you.
- It connects to the internet safely and keeps all your info private.

What We Learned: This project really taught us how important it is to pick the right parts, manage battery life, and design things with real people in mind, especially for health gadgets.

A Huge Thanks To...

We want to give a big shout-out to our awesome professors and the department for all their guidance and support throughout this whole capstone project!

Questions & Let's Chat!

Got any questions for us? We'd love to talk about how we built this, why we made certain choices, or what cool things this medication system could do next!